

Lab 1: Product and Supplier Management

Node.js • Express • MongoDB • Mongoose | Total: 10 points

1. Instructions

Use Node.js, Express, MongoDB, and Mongoose to build a REST API. The application must use a **.env** file containing:

PORT - the Express server port.

MONGODB_URI - the connection string for the **Lab1** database.

Do not hardcode these values in the source code. Create the **Lab1** database and import the provided data into its **suppliers** and **products** collections.

2. Create the Express Project - 0.5 points

Create the project using **Express Generator**. The project name must be your student ID (for example, **SE123456**).

3. Suppliers - 3.5 points

3.1 Supplier CRUD Operations - 2.5 points

Build the following operations for suppliers:

Requirement	Points
Get all suppliers	0.5
Get supplier details by ID	0.5
Create a supplier	0.5
Update a supplier by ID	0.5
Delete a supplier by ID	0.5

3.2 Virtual Population - 1 point

When retrieving supplier details, use **Mongoose Virtual Populate** to include the products belonging to that supplier in a virtual field named **products**. Do not store an array of product IDs in the supplier document.

4. Products - 6 points

4.1 Get All Products with Query Operations - 4 points

Build an endpoint that retrieves products and supports the query operations below. The filters must work together in one request.

Requirement	Points
Filter by category	0.5
Filter by a minimum and maximum price	0.5
Filter by tags	0.5
Search by product name	0.5
Sort by price	0.5
Paginate using page and limit; default limit is 3	1.5

4.2 Get Product Details by ID - 0.5 points

Retrieve a product by ID. Use **Mongoose Populate** to include its supplier's name in the response.

4.3 Create a Product - 0.5 points

Create a product that references a supplier.

4.4 Update a Product - 0.5 points

Update an existing product by ID.

4.5 Delete a Product - 0.5 points

Delete an existing product by ID.